Practice Over Prompts: What Predicts Learning Gains in an AI Math Tutor?

UC San Diego Computational Social Science
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Background & Motivation

Vocareum deploys an AI tutor for UC Math 3B (Calculus) — students can request help on any problem at any time.

The platform's working assumption: *more AI tutor usage → more learning.*

This analysis directly tests that assumption using **real behavioral data** from a live course (200+ UC students, Fall 2025).

Research Questions

RQ1 Does AI tutor interaction frequency predict student learning gains?

RQ2 What behavioral features actually drive score improvement?

RQ3 Are there topic-level difficulty patterns actionable for platform design?

Analysis Workflow

1 · Student-level models

Compared AI requests vs. practice volume as predictors of learning gain (OLS)

2 · Question-level model

Estimated topic difficulty with random student intercepts (lme4)

3 · Machine learning

Compared LR, Random Forest, and MLP; 5-fold CV

4 · Feature importance

Identified strongest predictors of learning gain

Limitations

Observational data — practice volume may reflect motivation or preparation, not only learning behavior.

Single course — findings are based on one Math 3B course and may not generalize to all settings.

Unmeasured confounds — outside study time and peer support were not captured.

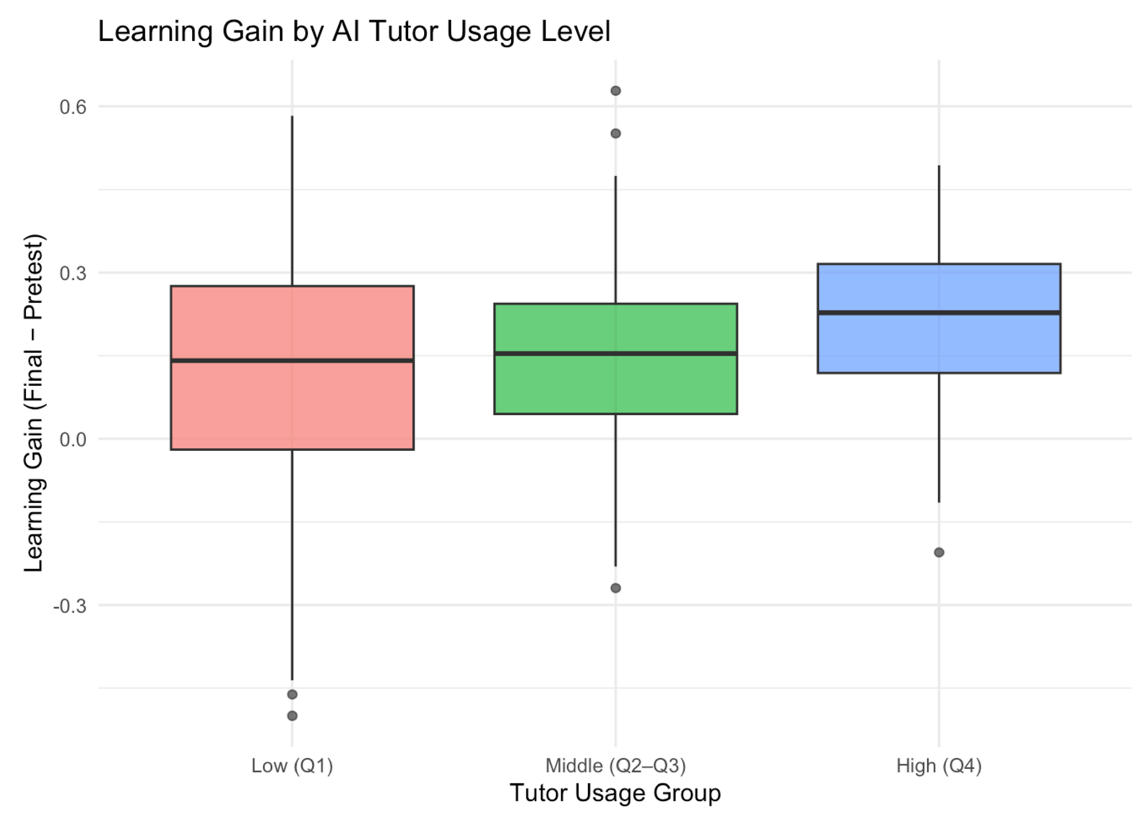
Request count ≠ quality — AI usage measures frequency, not whether the help was useful.

Dataset

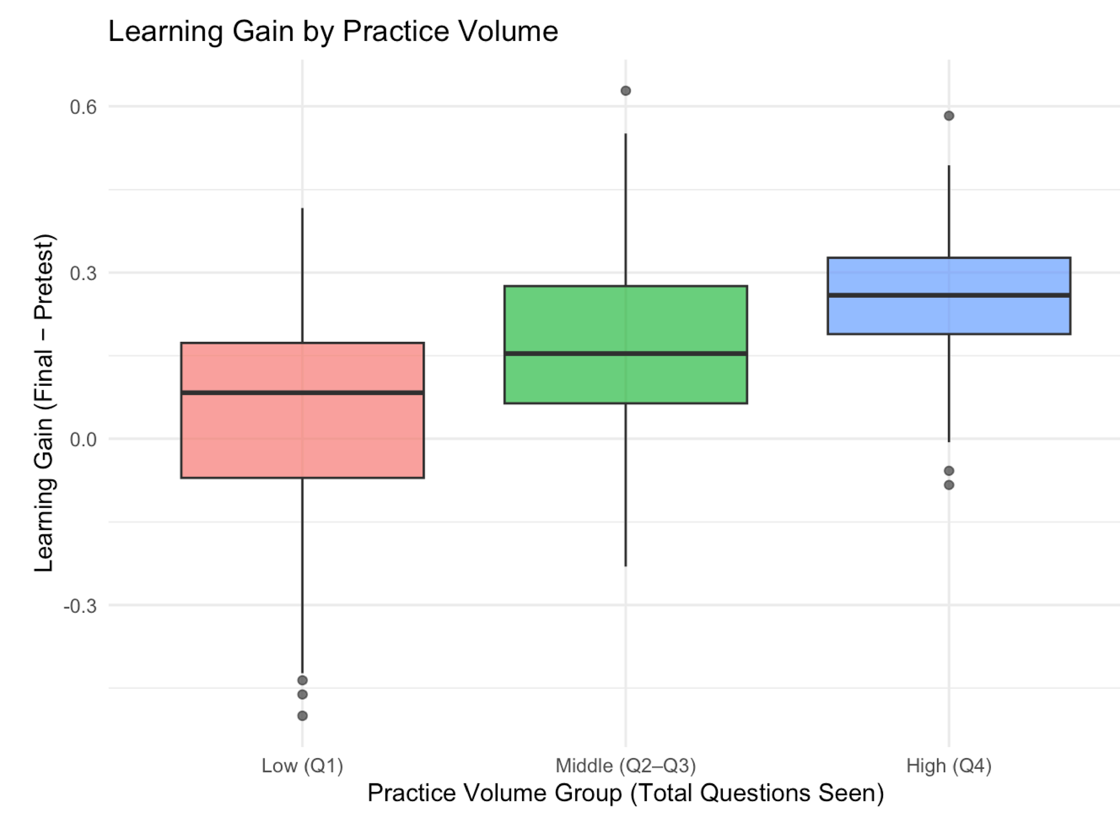
Item	Description
Platform	Vocareum Math 3B (Calculus)
Students	200+ UC students, Fall 2025
Outcome	Learning gain = Final − Pretest
Predictors	AI tutor requests, practice volume, accuracy, pretest
Privacy	*Raw data not released; all findings aggregated*

Key Finding: Practice Volume Predicts Gains — AI Tutor Requests Alone Do Not

Students who completed more practice problems showed higher learning gains — while higher AI tutor request frequency alone showed ***no consistent gain pattern*****.**

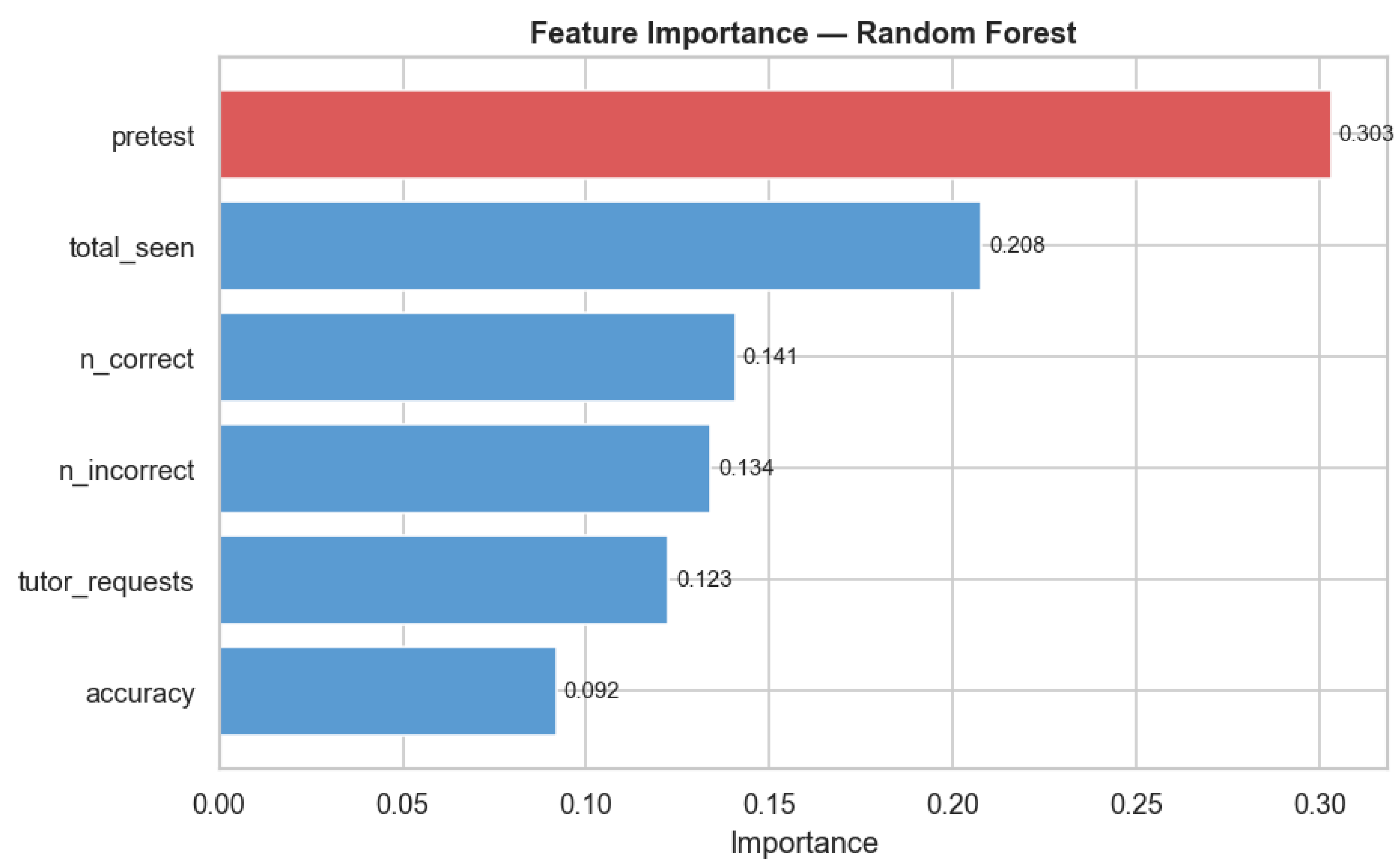


AI Tutor Usage groups (Low / Middle / High) show ***overlapping distributions*** *— no consistent learning gain trend*



Practice Volume groups show a ***clear upward gradient*** *— more questions attempted = consistently higher learning gains*

Feature Importance — Random Forest (Best Model, R^2 = 0.45)



Pretest score *dominates (0.303);* ***practice volume*** *(total_seen) ranks #2 (0.208);* ***AI tutor requests rank #5 of 6*** *(0.123) — below all practice-related metrics*

Pretest score was the strongest predictor — but practice volume ranked above AI tutor request count.

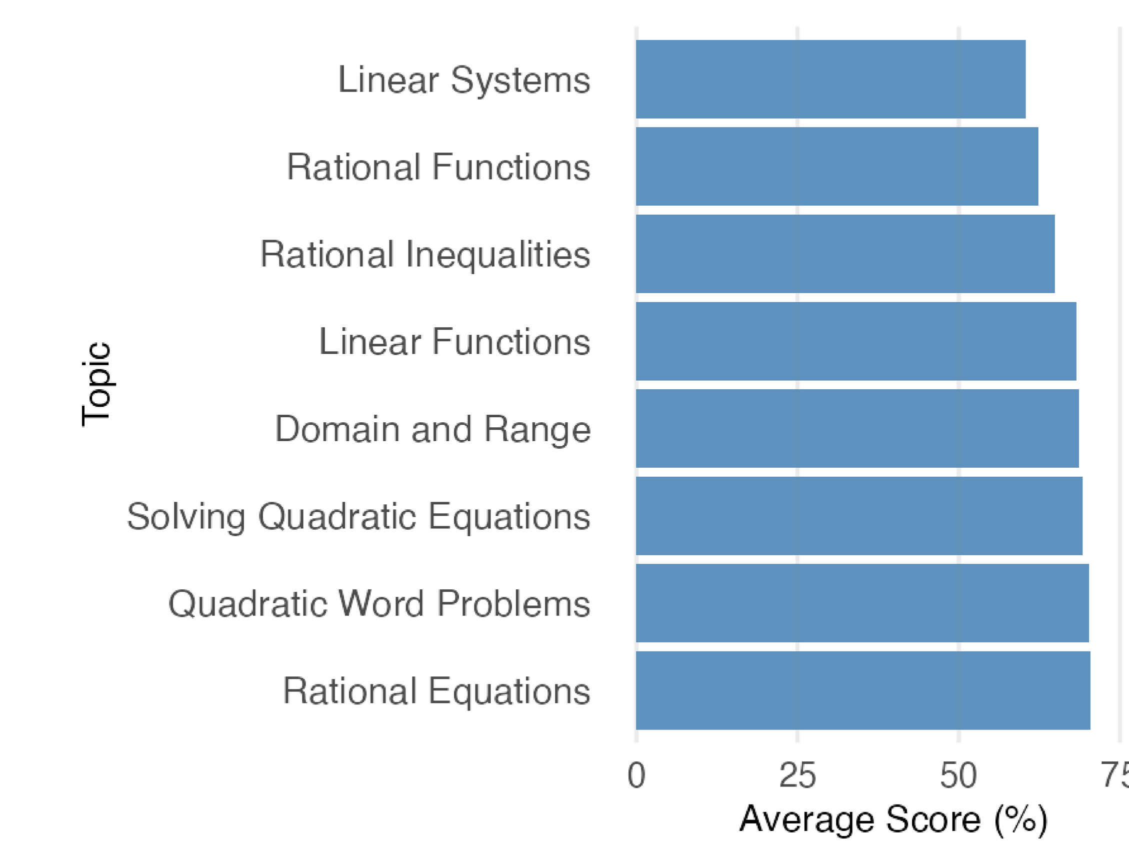
This suggests productive engagement matters more than simply asking the tutor more often.

Model Performance Summary

Model	Predictor(s)	R^2
OLS	AI Tutor Requests only	0.18
OLS	Practice Volume only	0.37
Lin. Regression	All features	0.42
Random Forest ★	**All features**	**0.45**
Neural Network	All features	0.34

R^2 = variance explained (0–1, higher is better). Random Forest ★ best predicts learning gains; AI tutor usage alone (R^2=0.18) is the weakest predictor.

Lowest-Scoring Topics: Where Support Is Needed



Linear Systems & Rational Functions consistently score lowest — top targets for instructional scaffolding

Conclusions & Future Directions

Practice completion — not AI tutor request frequency alone — was the stronger predictor of learning gains (RF R^2=0.45).

Future Directions

→ Validate findings across additional courses and student populations.

→ Test whether practice volume directly drives learning or reflects motivation.

→ Build dashboards to track practice completion and identify students needing support.

Product Implications for Vocareum

★ **Track practice completion, not only AI usage.** Practice volume was a stronger learning signal than tutor request count.

★ **Flag high-AI-use + low-improvement students.** This pattern may indicate difficulty rather than productive engagement.

★ **Target scaffolding to low-scoring topics.** Linear Systems and Rational Functions may need additional support.

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